Exact replication and screening of tropical finiteness certificates for central configurations

Felipe Santibáñez-Leal
CAOS Research program, `github.com/fsantibanezleal/CAOS_RESEARCH`

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Abstract

We report an exact, fully reproducible replication and screening study around Smale's 6th problem (finiteness of planar central configurations), with the following contributions: (i) an exact calibration of the Albouy–Chenciner mutual-distance equations in open tooling, including a machine rediscovery of their dimension-blindness (the regular tetrahedron coexists with the square in the equal-mass rhombus stratum of the 4-body distance system until the Cayley–Menger equation is adjoined, after which the square, with side a satisfying $a^3 = (4+\sqrt{2})/8$, the larger of the two positive roots of $32a^6 - 32a^3 + 7 = 0$, is the unique positive point of the stratum); (ii) a machine measurement of the *enrichment law*: adjoining the asymmetric (Roberts) equations and the energy–inertia relation renders the $n = 3$ distance ideal zero-dimensional directly, eliminating the coordinate-line components of the bare symmetric system; (iii) an independent, exact reproduction of the Jensen–Leykin tropical prevariety certificate of generic-mass finiteness for $n = 5$ (arXiv:2301.02305v2): both published f-vectors match digit for digit and the pointedness of the recession cone is verified by an independent exact parser; and (iv) a screening study over mass-valuation families and equation-system variants at $n = 4, 5$ which finds two new working valuation families beyond the two published, replicates generic-mass finiteness for $n = 4$ purely polyhedrally in seconds, and shows that removing the ten dependent symmetric Albouy–Chenciner equations destroys the certificate at every tested valuation: the "redundant" equations carry essential tropical information. Every comet of every screened cell is then decided exactly by a rational-arithmetic feasibility certificate, which upgrades the negative half of the screening from absence of a certificate to proof; adding further valid equations is measured separately, showing Dziobek's equations to be tropically active but not decisive at the hard case and the energy–inertia relation to be tropically inert; and two censuses that defeated our own engine are closed by an independent solver with exact cross-verification. All computations are exact (rational or integer arithmetic, or certified polyhedral computation); every stated number is reproducible from the released code and recorded hashes. The finiteness theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin, and Jensen–Leykin; the measurements in (i), (ii) and (iv) are, to our knowledge, new at this level of explicitness. Finiteness for five

bodies and all positive masses has since been claimed by three groups, one with a Lean formalization; we audit that formalization: its main theorem states exactly the classical claim, and none of its 2,751 stored and 85,756 generated modules contains a construct that bypasses the kernel.

1 Introduction

A *central configuration* (CC) of the Newtonian n-body problem with masses $m_1, \dots, m_n > 0$ is a configuration $x = (x_1, \dots, x_n)$, $x_i \in \mathbb{R}^d$, satisfying

$$\lambda\,(x_i - c) = \sum_{j \neq i} \frac{m_j (x_j - x_i)}{r_{ij}^3}, \qquad i = 1, \dots, n, \tag{1}$$

for some λ, where c is the center of mass and $r_{ij} = |x_i - x_j|$. Smale's 6th problem for the 21st century [4] asks, following Chazy [5] and Wintner [6], whether for every n and every choice of positive masses the number of *planar* central configurations, up to symmetry, is finite. The modern ladder is: $n = 4$ finite for all positive masses (Hampton–Moeckel [7]: computer-assisted BKK certificate, count in $[32, 8472]$); $n = 5$ finite except possibly on an explicit codimension-two subvariety of mass space (Albouy–Kaloshin [8]); the spatial 5-body Dziobek case handled generically by Moeckel [9] and explicitly by Hampton–Jensen [10]; $n = 6$ open, reduced by Chang–Chen [11] to twenty-four unsolved zw-diagram cases (their programme preprint states the count verbatim; we read it from the primary source); and, for *generic* masses, the recent purely polyhedral certificate of Jensen–Leykin [12], complete through $n = 5$. Rigorous equal-mass censuses for $n \leq 7$ are due to Moczurad–Zgliczyński [13]; with one negative mass a continuum exists (Roberts [15]), so positivity is essential.

This paper is a replication and screening study: it places the certificates of this ladder inside one open, reproducible toolchain, with exact artifacts, and then runs the experiments that the frontier papers themselves call for. Negative results are reported alongside positive ones, and every quantitative claim below is reproducible from the repository.

2 Exact calibration of the Albouy–Chenciner system (EXP-001)

Working in the mutual distances r_{ij} with the scale normalization $S_{ij} = r_{ij}^{-3} - 1$ ($S_{ii} = 0$), the symmetric Albouy–Chenciner (AC) equations [16, 7] are, after clearing denominators,

$$f_{ij} = \sum_{k=1}^{n} m_k \left[S_{ik}\,(r_{jk}^2 - r_{ik}^2 - r_{ij}^2) + S_{jk}\,(r_{ik}^2 - r_{jk}^2 - r_{ij}^2) \right] = 0. \tag{2}$$

Our exact implementation (sympy over $\mathbb{Q}$) reproduces, for four exact mass vectors, the classical $n = 3$ landscape: the Lagrange point $r_{12} = r_{13} = r_{23} = 1$ identically in symbolic masses, and exactly one positive collinear solution per ordering (Euler–Moulton; for equal masses the 2-middle chart value is $r_{12} = r_{23} = \sqrt[3]{90}/6$), with the degree-54 symbolic Euler eliminant persisted. For $n = 4$ each cleared AC polynomial has exactly 14 monomials, matching the count stated in [7].

The instructive result of the calibration concerns the equal-mass rhombus stratum. In the stratum $r_{12} = r_{23} = r_{34} = r_{14} = a$, $r_{13} = r_{24} = b$ of $n = 4$, the bare AC system has *two* positive solutions: the planar square, with $b^2 = 2a^2$ and $a^3 = (4 + \sqrt{2})/8$ (minimal polynomial

$32a^6 - 32a^3 + 7$), and the regular tetrahedron $a = b = 1$, whose bordered Cayley–Menger determinant equals $4 \neq 0$. The distance equations are dimension-blind, exactly as stated in [7]; planarity is restored only by adjoining the Cayley–Menger equation, after which the square is the unique positive point of the stratum. We record the scale-free invariant $J = UI^{1/2}/M^{5/2}$ at both anchors: $J_\square = \frac{1}{16} + \frac{\sqrt{2}}{8}$, $J_{\text{tet}} = \frac{3\sqrt{6}}{32}$.

Two toolchain findings of independent interest are covered by regression tests in the repository: `sympy`'s `solve_poly_system` silently returns incomplete solution lists on these systems (it missed the square, whose eliminant factor $4b^6 - 4b^3 - 7$ is even radical-solvable), and `is_positive` returns `None` on nested algebraic numbers, so naive filters silently drop valid solutions. Counting in our pipeline therefore uses univariate eliminants that are provably elements of the ideal (lex Gröbner elimination, or Stickelberger multiplication-matrix characteristic polynomials), complete real root isolation, and exact residual acceptance.

3 The enrichment law (EXP-002)

The bare symmetric system is never zero-dimensional: the line $r_{13} = r_{23} = 0$ lies in its variety for all masses (machine-verified symbolically), and saturation is expensive. Adjoining the $n(n-1)$ asymmetric equations $g_{ij} = \sum_k m_k S_{ik}(r_{jk}^2 - r_{ik}^2 - r_{ij}^2)$ (observed by G. Roberts; see [10, 12]) and the energy–inertia relation $U = MI$ changes the picture completely: for all four sample mass vectors the enriched ideal is zero-dimensional *directly* (Gröbner basis of size 24 in under a second each), and the full positive censuses, where decided, are exactly classical: four points (one equilateral, three collinear), with no spurious solutions. The planar rhombus census on the enriched system with Cayley–Menger adjoined yields the square alone. This is the affine-level instance of a pattern that recurs at every level of the subject: Hampton–Moeckel could not run their method on the six AC equations alone and required Dziobek's equations [7]; Hampton–Jensen kept the asymmetric equations to refine their tropical cones [10]; and Jensen–Leykin explicitly call for "different versions of the Albouy–Chenciner equations" at $n = 6$ [12]. Section 5 quantifies this pattern at the prevariety level.

4 Exact reproduction of the Jensen–Leykin $n = 5$ certificate (EXP-003)

Jensen–Leykin [12] prove generic-mass finiteness of planar CCs for $n \leq 5$ by a purely polyhedral computation: specialize the masses into the Puiseux field with chosen valuations, compute the tropical prevariety of the system (20 asymmetric AC + 10 symmetric AC + 5 planar Cayley–Menger polynomials for $n = 5$) with `gfan` 0.7, and verify that every connected component has a pointed recession cone; Bieri–Groves and the density of tropicalization fibers then yield finiteness for almost all masses.

We rebuilt `gfan` 0.7 from the author tarball (one two-line `#include <cstdint>` patch for gcc 13; tarball and binary SHA-256 recorded) and ran the paper's own command pipeline. Results: with valuations $(1, 3, 9, 27, 81)$ the prevariety f-vector is $(1506, 4744, 8586, 8787, 4652, 993)$, and among its 85 unbounded ray directions none has a positive coordinate (verified by our independent exact parser): the globally pointed certificate, exactly as published. With valuations $(1, 4, 9, 16, 25)$ the f-vector is $(3586, 12012, 18531, 15625, 7072, 1357)$, again matching [12] digit for digit. Each run took under six minutes of wall time on 30 threads (≈ 145 cpu-minutes). To

our knowledge this is the first published independent replication of the [12] computation; at the measured throughput, a single $n = 6$ attempt at the scale they report (≈ 100 cpu-days) costs about three wall-days on commodity hardware.

5 Valuation and equation-variant screening (EXP-004)

The certificate of [12] has two designed inputs: the mass valuations and the equation system. The paper publishes two working valuation choices for $n = 5$ and, for the inconclusive $n = 6$ attempt, asks explicitly for "more experimentation with different valuations or different versions of the Albouy–Chenciner equations". We screened both axes on a 16-cell grid: two systems (S1, the full set of asymmetric + symmetric AC equations + planar Cayley–Menger determinants; S2, the same with the $\binom{n}{2}$ dependent symmetric equations removed) crossed with six valuation families at $n = 5$, plus four cells at $n = 4$. Pointedness is decided per connected component (*comet*) by an exact certificate: a rational vector c with $c \cdot r < 0$ for every recession generator r, verified in rational arithmetic (a floating-point solver only proposes c).

Valuations at $n = 5$ (system S1)	comets	pointed	status
$(1, 3, 9, 27, 81)$ powers of 3	281	281	certificate (globally pointed)
$(1, 4, 9, 16, 25)$ squares	257	257	certificate
$(1, 2, 4, 8, 16)$ powers of 2	266	266	certificate (new)
$(2, 3, 5, 7, 11)$ primes	250	250	certificate (new)
$(0, 1, 2, 3, 4)$ arithmetic	181	180	no certificate (1 comet undecided)
$(1, 1, 9, 27, 81)$ repeated	211	210	no certificate (1 comet undecided)

Table 1: Comet-level screening at $n = 5$. The squares row reproduces the 257 components reported in [12], a quantity independent of the f-vector, which validates the component extraction as well as the polyhedral computation.

Three findings.

(a) Two new working valuation families. Powers of two and primes both certify at $n = 5$; before this screening, two families were known to work. Since the certificate is a statement about one point of the valuation space, each working family is an independent verification of generic finiteness for $n = 5$, and the enlarged pool is what makes an $n = 6$ attempt a design choice rather than a lottery. This matters operationally: powers of three reach t^{243} at $n = 6$ and exceed `gfan`'s machine-integer fast path (our first $n = 6$ attempt aborted on a detected overflow after six minutes), whereas powers of two reach only t^{32}.

(b) The dependent symmetric equations are essential. Every S2 cell has a strictly larger f-vector in every entry than its S1 counterpart and yields no certificate at any valuation, including the one where S1 is globally pointed; the single undecided comet grows from 71–95 recession generators in the S1 controls to 241–2224 in S2. The equations that are algebraically redundant (each symmetric equation is the sum of two asymmetric ones) are tropically load-bearing. This is the prevariety-level form of the affine phenomenon of Section 3, and it agrees with the historical record: [7] could not run their method on the six AC equations alone, and [10] kept the redundant family deliberately.

(c) $n = 4$ falls in seconds, and the hard case is the equal-valuation one. With valuations $(1, 3, 9, 27)$ and $(1, 2, 4, 8)$ the $n = 4$ prevariety is globally pointed (10 comets each; 4 and 2

seconds), a purely polyhedral replication of generic-mass finiteness for the four-body problem; arithmetic valuations $(0, 1, 2, 3)$ also certify (9 comets). The equal-valuation case $(0, 0, 0, 0)$ leaves a single undecided comet with 49 generators, matching the observation of [12] that this specialization is equivalent in difficulty to the original computation of [7].

5.1 Every comet decided exactly (EXP-007)

On the failing controls the heuristic of the screening returns neither certificate, so those cells read "no certificate" rather than "not pointed". An exact decision closes that gap. A cone is pointed exactly when the only nonnegative solution of $\sum_i \lambda_i g_i = 0$ is $\lambda = 0$, which is a single linear feasibility question; we decide it with a phase-I simplex in exact rational arithmetic (Bland's rule, so termination is guaranteed). Feasibility returns an explicit nonnegative zero combination, infeasibility returns a separating vector, and both are re-verified by exact substitution outside the decider.

All sixteen cells are decided in 33 seconds total, with zero certificate-verification failures. Every comet certified pointed by the screening is confirmed (no flips), and each failing control has exactly one comet that is provably *unpointed*: at $n = 5$ the arithmetic valuations $(0, 1, 2, 3, 4)$ (180 pointed, one unpointed) and the repeated valuations, and at $n = 4$ the equal valuations. Our data therefore independently confirms, rather than merely being consistent with, the failure of arithmetic valuations reported in [12], and localises it to a single component out of 181.

The logical direction matters: pointedness of every comet is a *sufficient* condition for dimension zero, so an unpointed comet proves only that this certificate route does not close, never that finiteness fails. The prevariety over-approximates the tropical variety.

5.2 Adding equations: what helps and what does not (EXP-008)

Since removing the redundant symmetric equations is catastrophic, the converse is worth measuring. At $n = 4$, where the planar stratum *is* the Dziobek stratum and Dziobek's equations are therefore available (they are what Hampton–Moeckel [7] needed when the Albouy–Chenciner equations alone defeated their method), we ran the baseline against three enrichments.

Two results run against intuition. First, adding equations does not always shrink the f-vector: it shrinks it at equal valuations (a 25 percent reduction) but *grows* it at powers of two and at arithmetic valuations. The prevariety as a set must not be confused with the subdivision that represents it: intersecting more tropical hypersurfaces can only shrink the set while refining the fan, so cone counts may rise. The invariant that behaves monotonically in our data is the comet count, which falls from 10 to 7 and from 9 to 6 when Dziobek is adjoined. Second, Dziobek's equations do not rescue the hard equal-valuation case: the prevariety shrinks by a quarter and its single comet stays provably unpointed. The enrichment that rescued the *algebraic* proof therefore does not rescue the *tropical* certificate at the same specialisation, which separates two roles that are easy to conflate.

By contrast the energy–inertia relation is tropically inert: it reproduces the baseline f-vector exactly at two of the three valuations. Dependence in the ideal predicts tropical relevance in neither direction, which is the sharper form of the lesson above, and it lets us drop that equation from expensive runs at no cost.

6 Censuses across two independent engines (EXP-006)

Two of the $n = 3$ censuses do not finish within a 900-second limit in our own engine. We recomputed them with msolve (Gröbner bases, rational univariate representation and certified real-root isolation), keeping our exact machinery as the verification layer: msolve returns rational isolating boxes, and containment of an exact algebraic point in a rational box is decided exactly. Both open cells close in under a second each, and for all four mass vectors the positive census is exactly the classical answer, one equilateral point and one collinear point per ordering, with every one of our independently computed exact points inside a distinct box and no unexplained boxes. Two implementations that share no code agree on every decided instance.

The same computation shows that the enriched planar $n = 4$ system does not admit an isolated census to compare with the published equal-mass class count: msolve reports the affine system to have dimension one, so there is no isolated census to compare. Diagnostics exclude the vanishing-distance loci as the cause, which leaves the reading the literature already acts on: the census belongs in the torus, where all mutual distances are invertible, which is precisely the setting of [7]. We record the count as untested rather than quoting the published value as if we had reproduced it.

Following that lesson, we attempted the $n = 4$ equal-mass census where it belongs, in the torus, by two independent exact formulations: the enriched planar system under a Rabinowitsch saturation of the distance product (seven unknowns) and the z-system of [7] (twelve unknowns, the mutual distances joined by the Dziobek variables). The exact square of Section 2 satisfies the first system, checked by exact residual substitution. Neither formulation produced any msolve output within a mechanically enforced time limit of 3600 seconds per route (EXP-009), so the published equal-mass counts remain untested by us. We note that the literature contains, to our knowledge, no direct algebraic solve of this census either: [7] bound the count by a mixed volume rather than solving, and the rigorous equal-mass counts of [13] come from interval methods. The measured cliff between $n = 3$ (both censuses close in under a second) and $n = 4$ (no output in an hour, twice) points counting towards bounds rather than solutions, and finiteness towards dimension computations, as follows.

Dimension of the $n = 4$ Dziobek variety (EXP-010). The object is the Dziobek variety of $n = 4$: the vanishing of the cleared Dziobek differences $S_{12}S_{34} = S_{13}S_{24} = S_{14}S_{23}$ together with the planar Cayley–Menger determinant, in the torus of the six mutual distances. Masses do not appear in this formulation, and generic finiteness corresponds to this variety having the dimension of the projectivized mass space, three. The instrument keeps every algebraic call zero-dimensional by construction: recorded random integer linear sections of codimension three (which should meet the variety in finitely many points, whose number is degree data) and codimension four (which should miss it), decided by msolve under caps, with a deterministic Gröbner staircase computation run separately. A validation test discriminates three exact anchors in one second: the exact square of Section 2 lies on the variety (certified by polynomial reduction modulo $8a^3 = 4 + \sqrt{2}$ written as two polynomial relations, no radical heuristics), the unit tetrahedron satisfies the Dziobek equations but is excluded by Cayley–Menger, and the 3-4-5 rectangle satisfies Cayley–Menger but violates the Dziobek equations.

The probes show an asymmetry that is itself the finding. Two independent codimension-four random sections (coefficients drawn uniformly from $[-10^6, 10^6]$ by a seeded generator, the draws archived verbatim) came back *empty* in one second each (msolve's dimension -1 output), which is probabilistic-exact support for the Dziobek variety having dimension at most three, exactly the

value the generic finiteness picture predicts and exactly the direction such arguments consume (what must be excluded is excess dimension). By contrast, both codimension-three census probes and a deterministic Gröbner staircase computation did not finish within their time limits (900 and 1800 seconds), so no degree data and no dimension lower bound beyond the exact witness were produced, and the census half of the instrument is not pursued at this size. Since only an a-priori degree ceiling is available (Bézout with the stripped degrees $9 \cdot 9 \cdot 6 = 486$), the emptiness result carries a failure estimate of order 10^{-3} per draw; it is evidence, not proof. For the upper-bound direction the natural tools are cheap emptiness probes and the partial-Gröbner device of Dias–Pan [14] (dimension bounds from the union of leading monomials of tractable subideal bases); certified witness sets serve the lower-bound direction. The next case is the spatial five-body Dziobek variety, following the proof shape of [14] with exact instruments.

The scaling test at $n = 5$ (EXP-011). At $n = 5$ the same instruments meet a computational wall. The object is the fifteen cleared Dziobek product differences over all quadruples of five bodies together with the spatial Cayley–Menger equation, in the torus of the ten mutual distances; the cut is an overvariety of the spatial Dziobek central configurations (degenerate strata satisfy the products trivially, as an exact point of the validation set shows: the unit-side, unit-slant bipyramid with $r_{45}^2 = 8/3$, whose membership reduces to rational arithmetic). The three-anchor validation passed in four seconds; every msolve section probe, including the codimension-five emptiness probes that cost one second at $n = 4$, capped at 300 seconds; and a run of the partial-Gröbner instrument (fifteen subideals, each one product plus Cayley–Menger plus the saturation, 120 seconds apiece under sympy) completed a single subideal, leaving a vacuous union bound. Nothing refuted the expected dimension four; nothing supported it either. We read the wall as an engine fact rather than a mathematical one: [14] report completing the identical leading-term pattern on an eleven-variable system in minutes under Singular, which motivates a change of Gröbner engine.

Engine comparison and the $n = 5$ dimension (EXP-012, EXP-013, EXP-014). Three further computations settle the engine question and reduce the $n = 5$ question to a single open decision. First, the wall is an engine effect: Singular over $\mathbb{Q}$ (version and binary hash recorded) completes twelve of the identical fifteen subideal jobs at 0.4–1 seconds each, against one completion near 100 seconds under sympy. An exact-reproduction control also shows that sympy's Gröbner output uses lexicographic ordering whatever order is requested, so the leading terms harvested in the EXP-011 run were lex leads of grevlex elements; that harvest gave only a vacuous bound, and no result depends on it. msolve's basis mode is modular, so it is used for screening only: an unlucky prime can enlarge a leading ideal and fake a *lower* dimension bound. Second, the union bound gives dimension at most 5, and the fifteen-product system without Cayley–Menger terminates completely in nine seconds, so its staircase dimension is the true Krull dimension: *exactly* 5, matching the rank-one parametrization count. The cost law is sharp: every computation without Cayley–Menger runs in seconds, every mix of the products with Cayley–Menger caps. Third, the incremental extension of the completed products basis by Cayley–Menger (whose normal form against that basis is nonzero, so the cut is a proper subvariety) also did not terminate within 1800 seconds. The deterministic results for the spatial five-body Dziobek cut are therefore: products variety of dimension exactly 5, cut of dimension at most 5, expected value 4 undecided. The remaining decision (does Cayley–Menger vanish identically on a top component?) calls for certified witness sets, and the cost law, measured twice, guides the work on the symmetric strata below.

7 The two-pair symmetric stratum of the six-body problem (EXP-015–018)

Dias–Pan [14] proved generic finiteness for the *cross* symmetric class of the planar six-body problem (four bodies on the reflection axis, one mirror pair). We study the neighbouring class, with two bodies on the axis and *two* mirror pairs, for which our literature searches found no published finiteness result. The exact quotient derivation gives nine distance variables; the relation $c_x^2 - c_s^2 = w_A w_B$ between the cross-side and same-side pair distances and the pair widths; a pair-equality lemma in closed form (the pair equations factor as the mass difference times $(q - v)$ times $(c_x^3 - c_s^3)$, forcing pair-equal masses whenever the two pairs sit at different heights, with the equal-heights sub-stratum explicitly excluded from scope); and, by a pure symmetry argument, a reduced system of *six* mass-linear equations.

Machine stages, all exact: the gauged shape variety has dimension 4 (one second per engine run, two independent dimension readings agreeing, the analogue of the cross case's $\dim E = 4$); the 6×4 mass-coefficient matrix has full rank 4 at two exact rational geometries (pure radical arithmetic, with no truncation); the equal-mass regular hexagon is verified to lie in the stratum through the reduced system and its dihedral symmetry degenerates the rank to exactly 3 (nonzero 3×3 minor in closed form, all 4×4 minors vanishing). Three Gröbner formulations of the rank-deficiency loci did not finish within their time limits, with the cost see-saw measured from both sides and a decisive modular feasibility screen; the remaining steps are therefore approached by written proofs with machine-verified identities.

This route settles a first case. The complete catalogue of the 84 nonzero 2×2 minors, computed with the inverse-cube differences as abstract symbols, contains four minors that factor into monomials times products of two such symbols; their Boolean analysis shows that rank below 2 forces one of two explicit branches (body 1, or body 2, at distance r_{12} from *both* pairs), and on each branch five further explicit degenerations are all violated by an exact 3-4-5 witness geometry. Hence the rank-at-most-1 locus meets the shape variety in dimension at most one: the $k = 2$ case of the fiber-dimension case chain is proved. The remaining cases are open, each with a proposed route ($k = 1$ by the same technique one level deeper; $k = 3$ by bordered minors inheriting the proven clean entries; $k = 4$ by a rank-4 central-configuration witness, the hexagon being excluded by its own symmetry). We claim no finiteness statement for this stratum until the chain closes; every identity above is in the repository with its machine verification.

8 The $n = 6$ attempt (EXP-005)

The screening of Section 5 turns the $n = 6$ question into a design problem with two dials. What we have measured is a *tooling* barrier, characterized on four attempts with `gfan` 0.7. On the machine-integer fast path the computation aborts within minutes on a *detected* overflow (the benign failure mode: `gfan` raises rather than returning a wrong answer), and it does so for powers of three and for powers of two alike, although the latter caps the valuation exponents at 32 instead of 243, so the failure is not a matter of valuation magnitude. On the arbitrary-precision path, the powers-of-two run died after fifteen wall-hours (about six cpu-days) on an internal assertion in the hypersurface-intersection code. `gfan` 0.7 therefore fails at $n = 6$ in both arithmetic modes, in two distinct ways, while the same binary reproduces the published $n = 5$ certificates exactly; [12] report their own $n = 6$ computation as completing (and inconclusive on pointedness), so their run differed in version, flags, or valuation. We rebuilt the pipeline on the

0.8beta release, validated it against our $n = 5$ control (exact f-vector agreement), and confirmed that it clears 0.7's first failure point. No 0.8beta run at $n = 6$ has completed (powers of two on the fast path and powers of three with arbitrary-precision integers, both with checkpointing), so the $n = 6$ case remains open here. A completed run would either extend generic-mass finiteness to $n = 6$ or localize the obstruction to specific comets, data that has not been published for any failed attempt.

9 The five-body case for all positive masses: claims and a formal audit (EXP-024)

In October 2026 three groups claimed that the planar five-body problem has finitely many central configurations up to similarity for every positive mass vector: Tomar [1], with a Lean 4 formalization and a supplement of certificates; Chen and Tong [2]; and Liu and Liu [3]. All three complete the Albouy–Kaloshin classification of singular sequences [8] on the exceptional mass set that [8] leaves open. The question of the previous sections for $n = 5$ is therefore settled, if these claims hold, and the priority is theirs.

We audited the formalization of [1] without rebuilding it. Its main theorem, $\forall m,\ (\forall i,\ 0 < m_i) \to \mathrm{FinitelyManyCC}\ m$, is built from a planar configuration $q : \{1, \dots, 5\} \to \mathbb{C}$ with distinct points, the equation $\lambda m_i (q_i - c) = \sum_{j \neq i} m_i m_j (q_i - q_j)/|q_i - q_j|^3$ with $\lambda \in \mathbb{R}$ and $c \in \mathbb{C}$, and similarity by every map $z \mapsto az + b$ or $z \mapsto a\bar{z} + b$, $a \neq 0$; finitely many central configurations means finitely many similarity classes. This is the classical statement: the sign of the force is absorbed by λ, the value $\lambda = 0$ admits no configuration for positive masses, and counting up to reflections changes counts, not finiteness. A comment-aware search of the 2,751 stored source files and a search of the 85,756 generated modules (each archive checked against its published SHA-256) for `sorry`, `admit`, axiom declarations, `native_decide`, `implemented_by`, `extern`, `ofReduceBool`, `unsafe` and `skipKernelTC` finds no occurrence in the formalization; the only hits are in the exporter used to feed independent kernels. The audit does not replace the kernel check, which the author reports for the full build and for two independent kernels; it shows that the theorem checked is the right one and that no source exempts itself from the check. The open case for this programme is $n = 6$.

10 Scope and data availability

The study is replication-first. The finiteness theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin, and Jensen–Leykin; the equal-mass censuses to Moczurad–Zgliczyński. Our measurements (the dimension-blindness anchors with their exact minimal polynomials and J-invariants; the enrichment law at affine and prevariety level; the screening tables) are machine facts about known objects, stated with their complete provenance. The censuses for two integer-separated mass vectors, which our sympy engine does not finish, are closed by an independent engine (`msolve`) with the sympy layer retained as exact verification, while the $n = 4$ torus censuses and the Dziobek census probes remain capped as reported above. No claim in this paper depends on a literature statement that we have not read in its primary source.
Code, artifacts and experiment records are available at
https://github.com/fsantibanezleal/CAOS_RESEARCH under
`problems/dynamical-systems/central-configurations/`. The identifiers EXP-001 to EXP-024 in the section titles and the text denote the experiment records in its `experiments/`

folder; each record holds the code, the artifacts and the result. Tool versions and binary hashes (`gfan`, `msolve`, Singular) are recorded with the experiments that use them.

References

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